NUNCHI

SERIES A · APRIL 2026

# The durable runtime for production agents.

The model is the same. The system is the variable.

CATEGORY Agent Coordination Plane

PRESENTING Will [Last Name], Founder

# Three forces. One window.

0 1

## Models commoditized

Frontier APIs differ less than 8% on coding tasks. The capability ceiling stopped moving. The infrastructure ceiling didn't.

*Sky-T1, Berkeley · Feb 2026*

0 2

## Protocols crystallizing

MCP, A2A, ERC-8004, x402 — the agent stack is converging on standards in a 6–12 month window. Whoever sits at coordination becomes the default integration.

*Linux Foundation · 97M monthly SDK downloads*

0 3

## Spend has arrived

Cursor at ≈$2B ARR. Claude Code at ≈$2.5B run rate. The buyers exist. The bills are real. Most pilots still don't reach production.

*Founder disclosures · McKinsey 2026*

*Sources detailed in appendix A1.*

# Agents broke reliability — again.

2 0 1 0 monoliths broke reliability. 2 0 2 0 microservices broke it again. **2 0 2 6** **agents.**

41–86%

of multi-agent deployments fail in production.

**79%** of those failures come from coordination, not from model capability.

***Klarna*** *reversed its all-AI customer service in May 2025. The agents could do the tasks. Klarna couldn't audit them.*

# Why this team builds this.

## Will [Last Name]

Founder & CEO

Prior

[Prior infra company]. Shipped [system] with [usage scale].

Track record

[1–2 line credibility statement on shipped agentic systems].

Why now

Built Roko alone over [N months] dogfooding the runtime against itself. The system has been self-hosting development for [N] weeks.

Bench

**Recruiting**

Founding eng (chain). Founding eng (ML routing). DevRel.

**Advisors**

[Advisor 1 — domain]. [Advisor 2 — domain]. [Advisor 3 — domain].

**Open offers**

[N] candidates in late-stage conversation, ex-[notable orgs].

*Bracketed text indicates founder fields to be filled before send.*

# Same outcome. Different infrastructure.

Today · ~200 LOC of orchestration

```
// LangGraph + retry + telemetry + auth wrapper
from langgraph.graph import StateGraph
from telemetry import wrap_otel
from auth import require_token
from cache import LRU
from router import ModelRouter
from gates import build_gate_pipeline

graph = StateGraph(AgentState)
graph.add_node('plan', plan_fn)
graph.add_node('route', route_fn)
graph.add_node('act', act_fn)
graph.add_node('verify', verify_fn)
...
// + retry policy, + cost tracking,
// + replay, + identity, + reputation,
// + audit trail, + cross-org sharing
```

With Nunchi · one command

```
$ nunchi audit deployment payments-svc

◆ identity    agent-7f3a · rep[code]=0.91
◆ route       haiku-3.5 (gate-tier 2)
◆ gates       compile · test · diff · sec
◆ knowledge   12 signals queried · 3 cited
◆ verify      8 / 8 steps · ZK receipt

✔ done in 6m41s · cost $1.42
✔ replay-id 0xa1b2c3d4...
```

# Coordination separated from execution.

*SDN's pattern. One layer up.*

**CONTROL PLANE · NUNCHI**

| identity | routing | knowledge | policy | replay |
|---|---|---|---|---|
| ERC-8004 | cascade · gates | HDC · substrate | language-agnostic | event log · ZK |

**DATA PLANE · AGENT EXECUTION**

| model calls | tools | memory | side effects |
|---|---|---|---|
| any provider | MCP + custom | session · vector | fs · http · shell |

*Ethane (Casado, Stanford 2007) reduced networks to dumb forwarding governed by centralized policy. Nunchi does the same for agents.*

# Observe. Decide. Enforce. Record.

*The loop the field said couldn't close on agents.*

01

## OBSERVE

Pulse · ephemeral

Every action emits structured events on the bus. Token, latency, tool, gate, decision — all captured at source.

→ 02

## DECIDE

Score · Route

Cascade routing chooses model and parameters per task. Knowledge substrate injects context. Reputation gates trust.

→ 03

## ENFORCE

Gate pipeline

11 language-agnostic gates: compile, test, diff, lint, security, semantic. Failures route back. Frontier calls are last resort.

→ 04

## RECORD

Signal · durable

Content-addressed, HDC-fingerprinted. Replayable. Citable cross-org. Becomes context for the next agent on the next task.

# Let me show you.

nunchi audit deployment payments-svc

# Forty-three dollars per task is what enterprises pay today.

*We get them to a buck forty-two. Three composable optimizations. Methodology disclosed.*

$44.86 → $1.42

naive baseline · per task

optimized · per task

*HAL benchmark, Princeton, ICLR 2026. Excludes prompt caching by design.*

how

**5×** prompt caching
**3×** cascade routing
**2×** gate pre-screening

**Honest read.** These three optimizations compose. The HAL baseline excludes caching by design — that's the reference point against which the field measures. Our $1.42 number includes Anthropic prompt caching priced per docs (verified), Princeton RouteLLM cascade, and our own gate pre-screening data. Each layer is reproducible with the **nunchi run --share** command.

# Three numbers.

*No GitHub stars. No download counts. No lines of code.*

## [N]

Design partners

Named logos in the appendix. Paying > signed DPA > LOI. No verbal-only pipeline on this slide.

*[Logo strip in appendix A8]*

## [N]M

Agent runs / month

Depth, not breadth. Agent runs and tool calls — the metric Casado's Investing in Orbit named as the thing GitHub stars cannot fake.

*Reproducible with nunchi run --share*

## [N]x

vs. comparable

[X] runs in [Y] months vs. [comp]'s same metric at the same stage. One velocity number, sharp comp, no list.

*Comp methodology in appendix A8*

*Numbers populated 24h before send. Slide is templated; bracketed values are placeholders only.*

# Layer cake. Not zero-sum.

*Identity below us. Execution beside us. Coordination is empty.*

| | identity | routing | gates | knowl. | replay | x-org | |
|---|---|---|---|---|---|---|---|
| **Nunchi** | ● | ● | ● | ● | ● | ● | *(this deck)* |
| Temporal | ○ | ◐ | ○ | ○ | ● | ○ | *durable execution* |
| Keycard | ● | ○ | ○ | ○ | ○ | ○ | *agent identity / auth* |
| LangGraph | ○ | ◐ | ○ | ◐ | ○ | ○ | *orchestration framework* |
| Nava | ◐ | ○ | ○ | ○ | ◐ | ◐ | *agent payment trust* |

● first-class ◐ partial / framework ○ absent

***We're customers of Keycard, not competitors.*** *Their per-transaction pricing literally requires a coordination layer above it.*

*Full grid (12 competitors, 11 capabilities) in appendix A5.*

# Software P&L now. Token economy when volume justifies it.

TODAY · EQUITY · NUNCHILABS INC.

**Open-source runtime**

Apache 2.0. Roko on GitHub. Infrastructure adoption first.

**Enterprise support contracts**

Temporal precedent. Annual SLAs, dedicated support, custom gate development.

**Managed cloud**

BSL with 4-year Apache conversion. Hosted runtime + chain access. Usage-priced.

18–24 MO · TOKEN · NUNCHI FOUNDATION

**Protocol fees**

Helium-hybrid burn-and-mint. Stablecoin-priced gas. NUNCHI required for identity staking, validator stake, governance.

**Deferred until volume justifies it**

Story Protocol template. Token graveyard (VIRTUAL −86%, ELIZAOS −99.98%, FET −94%) is the cautionary corpus.

**Dual-asset, separated**

Equity at NunchiLabs Inc. (Delaware C-corp). Token at Nunchi Foundation. Casado leads infra. Dixon leads chain.

# Twelve months from close.

| | | |
|---|---|---|
| 01 | Design partners | 8 named logos in production. 3 case studies published. |
| 02 | Cloud GA | Managed runtime live with billing, gates, replay, audit. SOC 2 Type II in progress. |
| 03 | Protocol depth | ERC-8004 reference implementation. MCP / A2A / x402 native integration shipped. |
| 04 | Chain testnet | Sovereign L1 testnet with Tokyo validator topology. HDC precompile benchmarked. |
| 05 | Series B trigger | Run-rate threshold met. Coordination is the noun a16z portfolio uses by default. |

*This is the inevitable architecture for agent coordination.*

A P P E N D I X

# Diligence material.

*Eight pages. Not presented live. Sent in the PDF.*

# How $44.86 became $1.42.

| Layer | Optimization | Multiplier | Source / verification |
|---|---|---|---|
| Baseline | HAL benchmark, naive | — | Princeton, ICLR 2026 (excludes caching by design) |
| 1. Cache | Anthropic prompt caching (priced per docs) | 5× | Anthropic API documentation, current pricing |
| 2. Route | RouteLLM cascade (haiku → sonnet → opus) | 3× | Princeton NLP, RouteLLM 2024 |
| 3. Gate | Pre-screening: compile, test, diff before frontier | 2× | Internal benchmark, reproducible |
| **Total** | **Composed, multiplicative** | **30×** | **$44.86 → $1.42 per HAL task** |

**Reproducibility.** Every layer of this stack runs from the open-source Roko CLI. **nunchi run --share** posts a public receipt of the dollar delta on any HAL task an evaluator wants to repeat.

# Eleven gates. Seven rungs. Language-agnostic.

| | | |
|---|---|---|
| 01 | compile | build / typecheck via toolchain (cargo · npm · go · pytest · etc.) |
| 02 | test | unit + integration via project's own test runner |
| 03 | diff | structural diff vs. base — bounded changes only |
| 04 | lint | style + complexity (rust-clippy · eslint · ruff · etc.) |
| 05 | security | secrets · dep scan · code-injection patterns |
| 06 | semantic | embedding-distance check vs. spec — does it solve the right problem |
| 07 | coverage | delta coverage threshold per change set |
| 08 | budget | token cost vs. configured envelope |
| 09 | latency | wall-clock vs. SLO |
| 10 | policy | org-defined rules (no PII · no external calls · etc.) |
| 11 | replay | deterministic re-run from event log produces identical receipt |

*Every gate calls the project's own toolchain via subprocess. Roko is not a Rust framework.*

# Identity is an asset, not a credential.

AGENT IDENTITY · ERC-8004

## agent-7f3a91

0xa1b2...c3d4 · token #4192 · transferable


Why this matters

**Static API keys** carry no behavioral history, can't be slashed, can't be queried cross-org.

**ERC-8004** carries on-chain reputation across seven domains. Slashable on policy violation. Queryable cross-org without disclosure.

**ZK-HDC** lets one agent prove it knows something without revealing what. Circom + Groth16. Sub-second proving on a laptop. ~250K gas to verify.

*ERC-8004 is the community standard created to address A2A's missing 'agent discovery and trust' layer.*

# Sovereign EVM L1. Specs frozen. Implementation deferred.

*We are not pretending the chain runs in production today. The simulator (mirage-rs) does. The mainnet is a 2027 milestone.*

| | |
|---|---|
| Consensus | Simplex (Chan & Pass, IACR 2023/463) |
| Block time | ~50 ms via co-located Tokyo validators (Hyperliquid topology) |
| Validators | 21-of-N HotStuff BFT, Tokyo data center co-location |
| Runtime | reth / revm fork with native precompiles (not Stylus) |
| HDC precompile | 0xA01 · ~400 gas top-K similarity over 10,240-bit binary vectors |
| Identity | ERC-8004 · transferable ERC-721 · 7-domain EMA reputation |
| Job market | ERC-8183 · RandomVRF, Blind Auction, Direct Hire |
| Knowledge | On-chain substrate with demurrage-based pruning |
| ZK | Circom + Groth16 · Hamming distance over committed hypervectors |
| Status | **Specs complete · 76 implementation items deferred Tier 6 · testnet 2027** |

# Twelve named competitors. Eleven capabilities.

| | ID | RT | GT | KN | RP | X-O | REP | SLA | ZK | OS | $ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Nunchi** | ● | ● | ● | ● | ● | ● | ● | ● | ● | ● | ● |
| Temporal | ○ | ◐ | ○ | ○ | ● | ○ | ○ | ● | ○ | ○ | ● |
| Keycard | ● | ○ | ○ | ○ | ○ | ○ | ◐ | ◐ | ○ | ○ | ● |
| LangGraph | ○ | ◐ | ○ | ◐ | ○ | ○ | ○ | ○ | ○ | ● | ◐ |
| LangChain | ○ | ◐ | ○ | ◐ | ○ | ○ | ○ | ○ | ○ | ● | ◐ |
| Nava | ◐ | ○ | ○ | ○ | ◐ | ◐ | ○ | ○ | ○ | ○ | ● |
| Capsule | ◐ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ◐ |
| t54 Labs | ◐ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ◐ |
| Tempo | ○ | ○ | ○ | ○ | ○ | ◐ | ○ | ○ | ○ | ○ | ● |
| 0G Labs | ○ | ○ | ○ | ◐ | ○ | ○ | ○ | ○ | ○ | ○ | ● |
| Olas | ◐ | ○ | ○ | ○ | ◐ | ◐ | ◐ | ○ | ○ | ● | ● |
| AIOS / OS-claimants | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ◐ | ◐ |

ID identity RT routing GT gates KN knowledge RP replay X-O cross-org REP reputation SLA enterprise SLA ZK zero-knowledge OS open source $ funding/revenue

# Five objections. One line each.

*Anticipating these is the signal. We do not assume they will all be raised.*

**Q1. "Coordination is a feature of LangGraph / Temporal, not a category."**

LangGraph is a framework — single-tenant, no shared identity, no cross-org reputation. Temporal owns 'did this code run.' Nunchi owns 'did the right agent, with the right memory, at the right price, with a receipt the counterparty can verify.' Frameworks compose into Nunchi. Nunchi does not compose into a framework.

**Q2. "You can't close the control loop on agents (Casado, April 2025)."**

We agree the model can't close it on itself. The runtime closes it. Observe (Pulse) → decide (Score) → enforce (Gate) → record (Signal) — every stage external to the model, every stage deterministic, every stage replayable. The loop closed when we made the runtime, not the model, the responsible party.

**Q3. "This sounds like Keycard, and Keycard already exists."**

Keycard answers 'is this agent allowed to call this tool right now.' We answer 'which agent should call which tool, when, on which model, with what context, at what cost.' Their per-transaction pricing requires a coordination layer above it. We're customers of Keycard, not competitors.

**Q4. "Why a chain at all? Why not just a managed service?"**

Today, none. The chain is deferred — Tier 6, 2027. Today's product is the open-source runtime plus enterprise support contracts plus the cloud. The chain becomes the durable moat when cross-org coordination volume justifies it. The runtime works without the chain. The chain doesn't ship until the volume is real.

**Q5. "Open-source runtime → commodified margins."**

Apache 2.0 runtime is the developer wedge. Margin sits in (a) the cloud (BSL with 4-year Apache conversion), (b) enterprise support SLAs (Temporal precedent — zero commercial product at Series A, $5B at Series D), and (c) the eventual protocol-fee economy when the chain ships.

# From identity ratios up to addressable agent spend.

## 82:1

Machine-to-human identity ratio

CyberArk / Entro Security 2025. Already lopsided. Trending toward 144:1.

## $18.7B

NHI market by 2030

Non-human identity. 11.9% CAGR. Growing into the gap Nunchi addresses natively.

## $2.5B

Single-vendor agent spend

Claude Code run rate, Q1 2026. Cursor at ~$2B ARR. The buyer side of the trade is real.

**How we size.** We don't quote $230B agent TAM. We quote the part of the trade we can actually capture: identity primitives ($18.7B NHI by 2030) + coordination spend (10–30× cost reduction times agent spend that already exists at $2.5B at one vendor). That's a credible SAM, not a slide-deck dream.

*Methodology and source citations on each number available in the Series A memo.*

# Who's in the room behind the room.

## TEAM

Will [Last] · Founder, CEO

[#2] · Founding eng (chain) · in late stage

[#3] · Founding eng (ML routing) · interviewing

[DevRel] · open req, ex-[org] · in pipeline

## ADVISORS

[Advisor 1] · ex-[notable infra company]

[Advisor 2] · author of [relevant paper / standard]

[Advisor 3] · operator at [scale infra]

[Advisor 4] · regulatory / policy

## DESIGN PARTNERS

[Logo 1] · paying · [usage scale]

[Logo 2] · signed DPA · in deployment

[Logo 3] · LOI · pilot scoped

[Logo 4–8] · early conversations

Hebbia · Harvey · Decagon · Sierra · Resolve.ai

*Bracketed values populated 24h before send. Logos arranged by stage of engagement (paying > DPA > LOI > pipeline).*